

STANDARD COLLEGE NTUNGAMO
S.5 SUBSIDIARY ICT
EOT 2 MARKING GUIDE
S850/1

SECTION A

ITEM 1

(a) Assessment of Mrs. Namukasa's Situation and ICT Tools Required

To successfully prepare for the Nyamunuka People's SACCO Annual General Meeting (AGM) at Kyamate Playground for over 20,000 members, Mrs. Namukasa requires a combination of specialized hardware and software tools tailored for large-scale outdoor visual display, digital data capture, media recording, and high-volume physical document generation.

High-Lumen Outdoor Digital Projector / LED Video Wall

An optical output display device capable of receiving video signals from a computer and displaying images onto a large screen. For an outdoor venue like Kyamate Playground in broad daylight, it requires an extremely high brightness rating (at least 6,000 to 10,000+ Lumens or an outdoor LED video screen) and Full HD resolution. Its purpose in this scenario is to display a clear, high-contrast visual summary of the manager's speech so that the large audience of 20,000 members can view it comfortably from a distance.

Computer Set

A desktop or laptop workstation consisting of a system unit, display screen, keyboard, and mouse. It should feature a high-performance multi-core processor (such as an Intel Core i5/i7), 16 GB of RAM, and a 512 GB Solid State Drive (SSD) running modern system software such as windows 11 professional. Its primary purpose is to act as the central workspace where Mrs. Namukasa processes attendance records, formats policy guidelines, and manages media files.

Barcode / Biometric Scanner (Digital Attendance System)

An electronic input device used to capture digitized data rapidly from member identity cards or fingerprints. Its purpose in this scenario is to digitally record the attendance of over 20,000 arriving SACCO members quickly and accurately, avoiding long queues and manual paper registration errors.

Digital Camera / Camcorder

An optical input capture device used to record high-resolution still images and digital video clips. Its purpose is to record high-quality photographic evidence and video footage of the AGM proceedings, speeches, and member interactions for the SACCO's historical archives and publicity.

High-Volume Heavy-Duty Multifunction Printer

An enterprise-grade output printing and scanning unit with high-speed printing capabilities (DPI resolution of at least 1200 × 1200) and automatic duplexing. Its purpose in this scenario is to rapidly produce 20,000 physical hard copies of membership certificates and policy guidelines for all attending members before printing deadlines.

Presentation Software (e.g., Microsoft PowerPoint 2024, Google Slides)

An application software used to create structured visual slide sequences. Its key features include pre-designed templates, high-contrast text styling, and multimedia embedding. Mrs. Namukasa will use it to structure and format the manager's speech into an engaging visual slideshow for outdoor projection.

Database / Spreadsheet Software (e.g., Microsoft Excel 2024, MS Access)

An application software providing interactive tables for structured data organization, storage, and analysis. It will be used to store, sort, and log the digital attendance data captured at the entrance gates.

Word Processing Software (e.g., Microsoft Word 2024, Google Docs)

A software application providing advanced text formatting, page layout design, and mail merge capabilities. Mrs. Namukasa will use it to format, style, and print the bulk membership certificates and policy manuals cleanly.

(b) Analysis and Addressing Computer Behavior & USB Issues

Computer Behavior Analysis: When Mrs. Namukasa connects her USB flash drive, the operating system displays a "USB Device Not Recognized" error. This indicates a failure in communication between the computer's operating system and the flash drive controller. Common root causes include corrupted or outdated USB port drivers, file system corruption on the flash drive, physical damage/dirt inside the USB connector, insufficient power delivery to the USB port, or static/port initialization glitches inside the system unit.

Step-by-Step Troubleshooting of Current USB Issue:

Perform a Physical Inspection and Clean Ports:

Unplug the USB drive and visually inspect its metal connector for dirt, dust, or physical damage. Clean the connector gently and reinsert it firmly into a direct rear motherboard USB port (if using a desktop) rather than front hub extensions.

Restart the Computer (Hard Power Cycle):

Save all open work, shut down the computer completely, and unplug the main power cable for 15–20 seconds to drain lingering static electricity. Reconnect power and turn the system back on to reset the USB host controllers.

Reinstall/Update USB Controllers in Device Manager:

Right-click the Start menu, open Device Manager, and expand the Universal Serial Bus controllers category. Right-click the unrecognized device (or generic USB Hub), select Uninstall device, and restart the computer. Windows will automatically reinstall the USB drivers upon reboot.

Test the USB Drive on an Alternative Computer:

Plug the USB flash drive into another working computer. If it opens successfully on the second machine, the original computer's drivers or ports are at fault. If it fails on the second computer, the flash drive itself has hardware or file system corruption.

Run Disk Management / Command Prompt Repair:

If detected without a drive letter, open Disk Management and assign a new drive letter. Alternatively, run `chkdsk X: /f` (where X is the flash drive letter) via Command Prompt to fix file system errors.

Addressing Future Challenges:

- **Adopt Cloud Storage Backups:** Save all speech drafts, policy manuals, and certificate templates to secure cloud storage platforms (such as Google Drive or Microsoft OneDrive). This ensures instant file retrieval from any computer if a physical flash drive fails.
- **Maintain Multiple Storage Redundancies:** Keep secondary copies of critical files on at least two separate, tested physical storage drives (such as an external SSD or a secondary flash drive).
- **Implement Version Control and Proper Naming:** Save documents using clear, descriptive names with version tags (e.g., `SACCO_Policy_Guideline_2026_v1.docx`) to prevent file loss or accidental overwriting.
- **Install and Update Antivirus Software:** Keep reliable security software running to prevent malicious software from corrupting USB file systems or blocking USB port communications.

SECTION B

ITEM 2

(a) Distinguishing Devices in Box A and Box B, and Primary Functions of Box B Items

Distinction Between Box A and Box B Components:

- **Box A (Internal hardware Components):** Contains hardware components that are installed directly inside the system unit casing. They form the internal core responsible for system power, processing instructions, temporary memory execution, and internal data storage.
- **Box B (External Peripheral Devices):** Contains external hardware components that connect to the system unit via external ports (such as USB, HDMI, audio jacks). They act as human-interface, data collection, visual feedback, or physical output peripherals.

Primary Functions of Items in Box B:

- **Monitor:** Acts as the primary visual display output device that provides real-time user interface feedback, displaying operating system graphics, software applications, and video signals.
- **Keyboard:** An input device used to enter alphanumeric text, symbols, commands, and navigational shortcuts into the computer system.
- **Mouse:** A pointing input device used to navigate the graphical user interface (GUI), control the cursor, select items, and execute commands via clicks.
- **Barcode Reader:** An optical input device that scans and decodes barcoded data printed on labels or products, transferring the encoded digital sequence instantly to the computer.
- **Printer:** An output device that converts digital text and graphical documents into physical hard copies on paper.
- **Projector:** An optical display output device that receives video signals from the computer and projects expanded images onto a large screen or wall for viewing by a large audience.
- **External Speakers:** Audio output devices that receive analog or digital sound signals from the computer's sound card and translate them into audible sound waves.

(b) Categorization and Safe Connection Guide for Box B Devices

Categorization of Box B Devices:

Device Category	Devices Included
Input Devices	Keyboard, Mouse, Barcode Reader
Output Devices	Monitor, Printer, Projector, External Speakers

Step-by-Step Guide to Safely Connect Box B Peripherals to the System Unit:

Unpack and Prepare the Workspace:

Unbox all peripheral devices and place the system unit, monitor, keyboard, and mouse on a clean, dry, stable surface. Ensure the master power switch on the Power Supply Unit (PSU) at the back of the system unit is set to the OFF (0) position, and all power cables are completely unplugged from wall sockets.

Connect the Display Devices (Monitor and Projector):

Connect one end of an HDMI or DisplayPort cable to the video port on the monitor and the other end directly into the dedicated graphics card port (or motherboard video port) on the back of the system unit. For the projector, connect its HDMI cable into an available secondary video output port on the back of the system unit.

Connect Input Peripherals (Keyboard, Mouse, and Barcode Reader):

Plug the USB cables from the keyboard and mouse into the designated USB ports on the rear panel of the system unit. Plug the barcode reader's USB cable into an available rear or front USB port. Note: Plugging input devices into rear USB ports ensures direct motherboard contact and prevents driver responsiveness glitches.

Connect Audio and Printing Peripherals (Speakers and Printer):

Plug the 3.5mm audio jack from the external speakers into the green audio output line-out port on the back of the system unit (and connect its USB/power cable if required). Connect the USB data cable from the printer into an available USB port on the system unit.

Connect Power Cables to an Uninterruptible Power Supply (UPS):

Plug the individual power cords for the system unit, monitor, printer, and projector firmly into their respective power inlets. Plug all these power cables into an Uninterruptible Power Supply (UPS) or surge protector—never directly into wall outlets—to protect the lab equipment against power surges.

Perform the Sequential Daily Power-On Procedure:

Turn on the main wall switch, then turn on the UPS. Switch on external devices (monitor, printer, projector, speakers) first. Finally, press the main power button on the front of the system unit tower to allow the computer to detect all connected devices during its Power-On Self-Test (POST).

ITEM 3

(a) Media File Size Analysis, Calculations, and Storage Device Selection

To determine the total storage capacity required to hold the entire media collection from the Inter-House Sports and Drama Competition, all file sizes are calculated and converted into Gigabytes (GB) using standard binary computer storage conversions (1 GB = 1,024 MB and 1 MB = 1,024 KB).

Total Size of High-Resolution Photos

- **Number of photos:** 1,200 files
- **Average size per photo:** 5 MB
- **Total size in Megabytes (MB):**

$$\text{Total Size (MB)} = 1,200 \times 5 \text{ MB} = 6,000 \text{ MB}$$

- **Conversion to Gigabytes (GB):**

$$\text{Total Photo Size (GB)} = 6,000 \text{ MB} \div 1,024 \text{ MB/GB} = \mathbf{5.859 \text{ GB}}$$

Total Size of High-Definition Video Clips

- **Number of video clips:** 10 clips
- **Average size per video clip:** 1.5 GB

- **Total size in Gigabytes (GB):**

$$\text{Total Video Size (GB)} = 10 \times 1.5 \text{ GB} = \mathbf{15.000 \text{ GB}}$$

Total Storage Capacity Required

- **Total size of all media files:**

$$\text{Total Size} = \text{Photo Size (GB)} + \text{Video Size (GB)}$$

$$\text{Total Size} = 5.859 \text{ GB} + 15.000 \text{ GB} = \mathbf{20.859 \text{ GB}}$$

Evaluation of Available Secondary Storage Devices in the Store

4 x Flash Disks (8 GB capacity each):

Total Combined Capacity: $4 \times 8 \text{ GB} = 32 \text{ GB}$.

Evaluation: Although the combined capacity is 32 GB, a single 8 GB flash disk cannot hold the total collection (20.859 GB). Splitting the media collection across 3 to 4 separate flash drives creates fragmentations, makes archival tedious, and increases the risk of file misplacement.

2 x Standard DVDs (4.7 GB capacity each):

Total Combined Capacity: $2 \times 4.7 \text{ GB} = 9.4 \text{ GB}$.

Evaluation: Insufficient. The 9.4 GB total capacity is far below the required 20.859 GB.

1 x External SSD (64 GB capacity):

Total Capacity: 64 GB.

Evaluation: Fully sufficient. A single 64 GB External SSD can hold the entire 20.859 GB media collection in one central location while leaving approximately **43.141 GB of free space** for future archives.

Final Recommendation & Justification

The **1 x External SSD (64 GB capacity)** is the **most suitable storage device** available in the store.

Justification:

- **Capacity:** It easily accommodates the entire 20.859 GB media collection on a single continuous drive without needing to split files across multiple media.
- **Performance & Speed:** SSDs offer significantly faster read/write data transfer speeds compared to flash disks and optical DVDs, saving substantial time when copying high-definition video files.
- **Durability & Safety:** SSDs have no moving mechanical parts, making them resistant to shock, physical drops, and magnetic corruption, thereby guaranteeing long-term archival safety for the District Education Officer and the school media club.

(b) Step-by-Step Procedure for Safe Data Transfer and Device Disconnection

Connect the External SSD to the Computer:

Locate a high-speed USB 3.0 / USB-C port on the rear panel of the school desktop computer. Plug the External SSD cable firmly into the port. Open File Explorer (Windows Key + E) and confirm that the External SSD is recognized as a removable drive (e.g., Drive (E:) or Drive (F:)).

Create a Structured Target Directory:

Double-click to open the External SSD. Right-click inside the drive window, select New > Folder, and name the directory Mubende_HS_InterHouse_Competition_2026. Inside this folder, create two subfolders named 01_HighRes_Photos and 02_HD_Videos.

Locate and Copy the Media Files:

Navigate to the local hard drive folder containing the competition media. Select the 1,200 photo files, right-click, and select Copy (or press Ctrl + C). Navigate to the 01_HighRes_Photos subfolder on the External SSD, right-click, and select Paste (or press Ctrl + V). Repeat this process for the 10 HD video clips into the 02_HD_Videos subfolder.

Verify File Integrity and Capacity:

Once the transfer progress bar completes, right-click the root folder on the External SSD, select Properties, and verify that the copied size matches 20.859 GB with all 1,210 items present. Open a sample photo and video file to confirm they play smoothly without corruption.

Safely Disconnect (Eject) the External SSD:

Close all open folders and media player applications accessing the External SSD. Go to the System Tray on the bottom right corner of the Windows taskbar, click the Safely Remove Hardware and Eject Media icon, and click Eject External SSD. Wait until the operating system displays the notification reading "*Safe to Remove Hardware*", then gently unplug the USB cable from the computer tower.

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